

FinSight

Enterprise Credit Risk Analytics & Automated Loan Underwriting Engine

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Problem Statement

Financial institutions evaluate large volumes of loan applications involving credit bureau data, income and transaction records, and repayment history. Manual credit risk assessment is slow, inconsistent across underwriters, difficult to scale, and prone to bias, while lenders must still meet strict regulatory and compliance requirements.

The core problem is the lack of a unified, data-driven system that can score applicant risk consistently, explain its decisions, and apply lending policy automatically — instead of relying on fragmented spreadsheets, manual checks, and disconnected scoring tools.

“FinSight” is developed to solve this problem by building an end-to-end credit risk analytics pipeline and an automated loan underwriting engine that scores applicants, explains the score, and issues a consistent, auditable lending decision.

Real-World Relevance

Credit risk scoring and loan underwriting sit at the core of banking, NBFCs, credit unions, and fintech lending platforms. Every disbursed loan carries default risk, and manually reviewing every application does not scale as loan volumes grow.

FinSight-style systems help banks and NBFCs approve creditworthy applicants faster, reduce non-performing assets by catching high-risk profiles early, and maintain audit trails required by regulators such as the RBI and Basel frameworks. Fintech lenders can use similar engines to expand access to credit for thin-file or first-time borrowers using alternative data, while keeping decisions explainable and consistent across every branch or channel.

Motivation for Selecting the Problem

Credit risk and underwriting is one of the highest-stakes applications of data science — a wrong decision either loses revenue (rejecting a good borrower) or creates a bad debt (approving a risky one), so the models must be both accurate and explainable.

This project was chosen to apply core data science and machine learning techniques — data cleaning, feature engineering, classification modelling, and model explainability — to a real, high-impact financial problem, while also exploring how automated decision engines and business rule layers are built around a model in a regulated industry.

The goal is to build a tool that goes beyond a single risk score — turning raw applicant and credit data into a transparent, automated lending decision — with the flexibility to extend across loan products, portfolios, and lending policies.

Objectives

- To build a data pipeline that ingests and cleans applicant financial, credit bureau, and transaction data.
- To engineer risk-relevant features such as income stability, debt-to-income ratio, and repayment history.
- To develop and compare credit risk scoring models (e.g., Logistic Regression, Random Forest, XGBoost).
- To design an automated underwriting engine that applies scoring output and lending policy rules to reach an approve / reject / refer decision.
- To integrate model explainability (e.g., SHAP) so every decision can be justified for compliance and audit.
- To build an interactive dashboard for risk analysts to monitor portfolio-level risk and underwriting outcomes.

Expected Outcomes

- A cleaned, feature-engineered credit and loan applicant dataset ready for modelling.
- A trained credit risk scoring model with evaluated performance (AUC-ROC, KS-statistic, precision/recall).
- An automated underwriting engine issuing consistent approve / reject / refer decisions.
- Explainability output showing the key factors behind each individual decision.
- An interactive dashboard for monitoring portfolio risk and underwriting activity.

Portfolio Risk Dashboard

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Avg. Risk Score

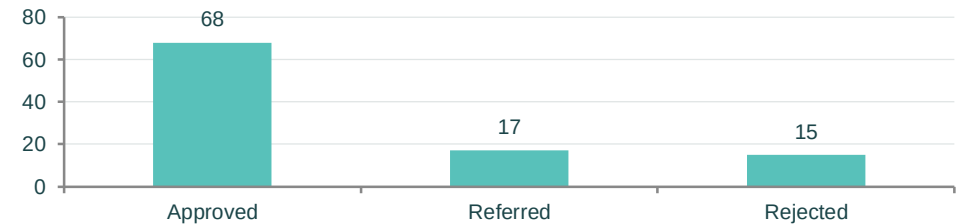
68%

Approval Rate

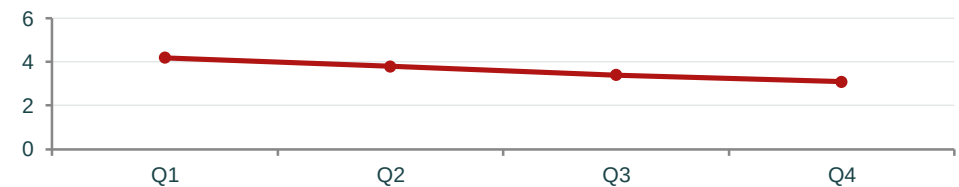
3.1%

Default Rate

Decision Outcome Mix (%)



Default Rate Trend



Literature Survey / Existing Systems

System / Approach	How It Works	Key Limitation
Traditional bureau credit scores (e.g., FICO-style)	Statistical score from bureau history and repayment behaviour	Ignores income, transactions, and alternative data; not portfolio-configurable
Manual underwriting review	Loan officer manually reviews documents and history per application	Slow, inconsistent across reviewers, difficult to audit or scale
Rule-based loan origination systems	Fixed eligibility rules (income, tenure, existing debt) applied in software	Rigid; cannot adapt risk cut-offs from data or explain a rejected case
Black-box ML credit models	Machine learning model predicts probability of default	High accuracy but low transparency; hard to justify to regulators or applicants

Limitations / Gaps

- Traditional credit scores rely on a narrow set of bureau variables and miss income or transaction signals.
- Manual underwriting is slow, subjective, and difficult to keep consistent across reviewers and branches.
- Rule-based origination systems cannot adapt cut-offs or weightings from data as portfolios evolve.
- High-accuracy ML models are often “black boxes,” making it hard to justify a rejection to a regulator or applicant.
- Most existing tools handle only part of the workflow — scoring, or rules, or reporting — not the full pipeline in one system.

How FinSight Addresses the Gap

FinSight combines credit bureau data, income and transaction history, and repayment behaviour into a single feature set— going beyond the narrow inputs used by traditional bureau scores.

The scoring model output feeds directly into an automated underwriting engine, so applications move from raw data to a decision without manual, reviewer-by-reviewer handling, keeping outcomes fast and consistent across every branch or channel.

Every decision is paired with an explainability report (key contributing factors) so approvals, rejections, and refer-for-review cases can be justified to auditors, regulators, and applicants — addressing the transparency gap in black-box ML models — while a configurable rule layer still lets risk teams adjust policy without retraining the model.

Thank You

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